

Direct Preference Optimization

DATA 443/643 · Week 9, note 9.3

Week 9

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Ideas from Stanford CS224N 2025 L10 (DPO: write r from π ; $Z(x)$ cancels). Original slides; that course is not copied.

Lecture plan

- 1 Skip the RM and the PPO loop
- 2 Closed-form r from the policy
- 3 Init loss is still $\log 2$
- 4 Offline: no new y unless you collect them

Key goal: the frozen reference is the leash. Dropping it is not DPO.

First time: what / why / formula

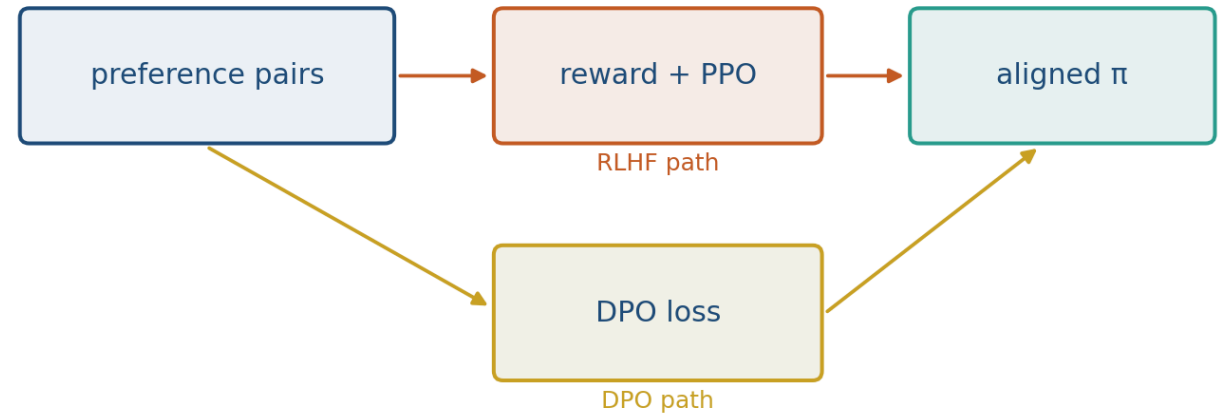
- What / why. Under Bradley–Terry, the optimal reward is $r = \beta \log(\pi / \pi_{\text{ref}}) + c(x)$. Plug into the logistic loss; r disappears. Train π_θ offline. RLHF's PPO stage is heavy. DPO is a classification loss on logits of y_w and y_l .
- Architecture. Policy + frozen reference. No critic, no sampling-in-the-loop required (you need logprobs of both completions).
- Formula. $\mathcal{L}_{\text{DPO}} = -\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)$.
- Tradeoffs. + Offline, no RM. – Still needs pairs and a reference; can overfit; not magic if the pairs are noisy.

How. Lab 9: toy scalar logits. Start: $\pi = \pi_{\text{ref}} \Rightarrow \text{loss } \log 2$.

Same pairs, shorter stack

- The optimal r is a log-ratio plus $Z(x)$.
- $Z(x)$ cancels in a pairwise difference.
- KTO and IPO exist. This hour is DPO.

DPO fits the same pairwise data without an explicit reward model



The implicit reward

$$r(x, y) = \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)}$$

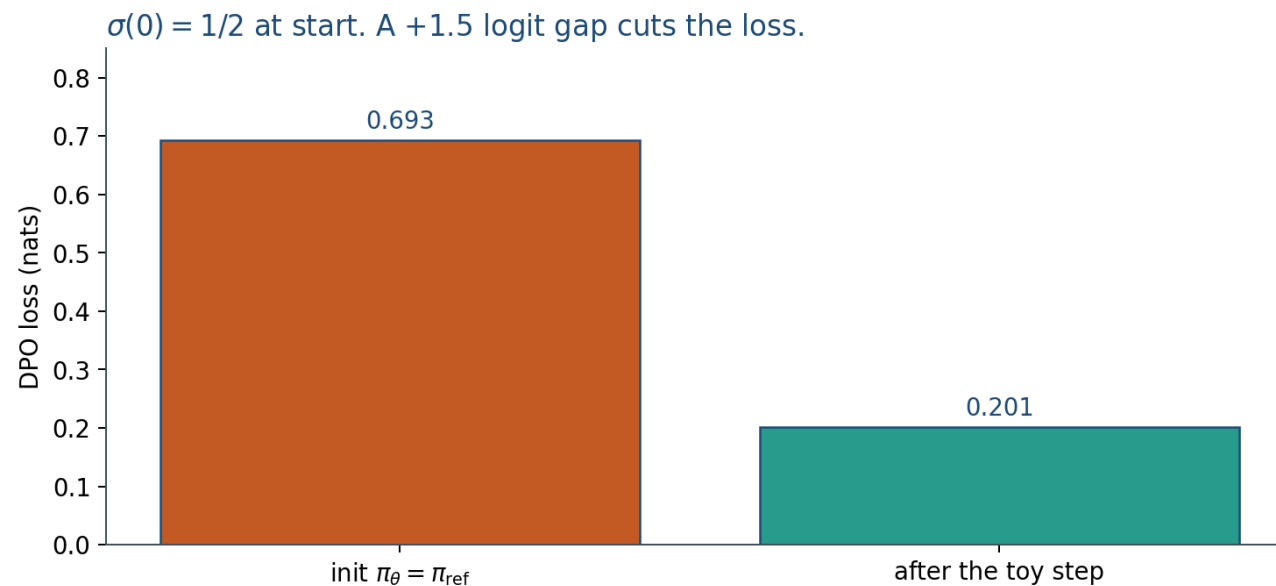
Plug into Bradley–Terry. The $\mathcal{R}$ head disappears.

If $\pi_{\theta} = \pi_{\text{ref}}$, the argument of σ is 0.

Trainable object is π_{θ} . The reference stays frozen.

A toy step

- Init: $\mathcal{L} = \log 2 \approx 0.693$.
- Winner log-ratio 0.5, loser -1.0 , gap 1.5.
- $\sigma(1.5) \approx 0.818$, loss ≈ 0.201 .



What the pair file cannot do

PPO

Samples new y from the current policy. Can explore replies that were never labeled.

Offline DPO

Fixed pairs. If SFT never wrote a good y_w , DPO cannot invent it.

What to take from this lecture

Init

$\sigma(0)=1/2$ is the start, not a bug.

Report

β , the reference checkpoint, held-out preference accuracy.

Lab 9

Toy logits. Not an LLM PPO run.